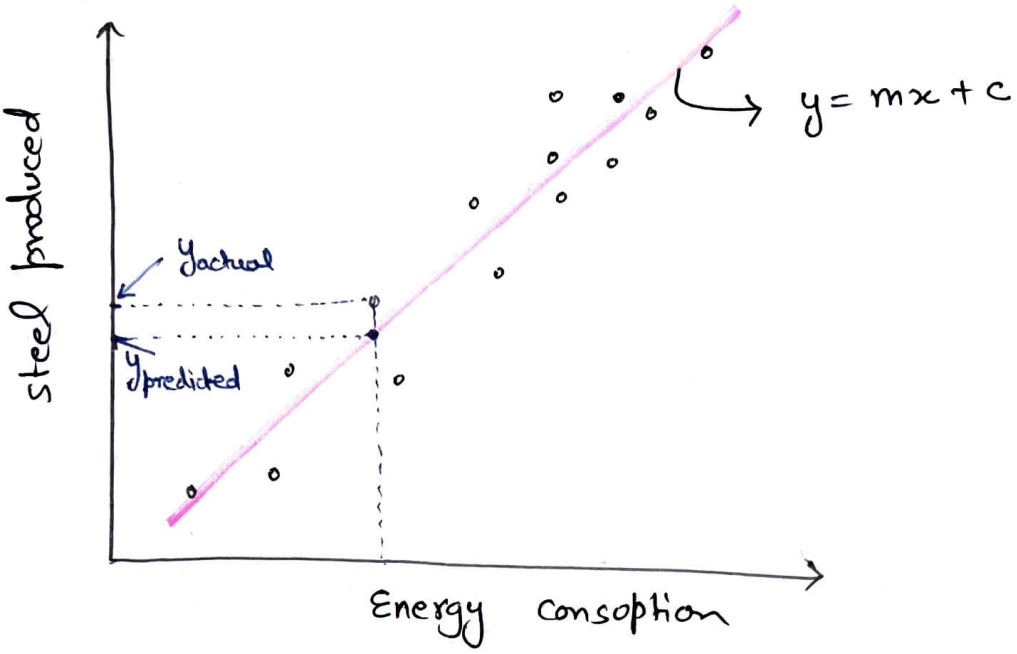
Least square Regression

Consider the problem where we want to predict the "steel production" based on "energy consumption"



We can solve this problem by fitting a linear equation of the form

$$F(x) = mx + c$$

where

$m \rightarrow$ slope of the line

$c \rightarrow$ intercept on the y-axis

This estimated equation may not perfectly fit all the data point, hence there always be some error

$$Error = | y_{actual} - y_{predicted} |$$

So, the estimated linear function "WILL BE THE BEST FIT, IF SUM OF ALL ERROR IS MINIMUM"

$$\sum_{i} Error_{Total} = \sum_{i=1}^{n} | y_{actual} - y_{predicted} |$$

where $n \rightarrow$ total number of samples in data

So we can assume this as a minimisation problem

$$\min E_{Total} = \min \sum_{i=1}^{n} (y - \hat{y})^2$$

since $\hat{y} = mx + c$, substituting its value

$$\min E_{Total} = \min \sum_{i=1}^{n} (y - (mx + c))^2$$

m and c are the unknowns we want to minimize the function for

using partial order derivative

$$\frac{\partial E}{\partial m} = \sum_{i=1}^{n} \frac{\partial E}{\partial m} (y - (mx + c))^2 = 0$$

Removing limits of summation for now

$$\sum * 2 (y - (mx + c))(-x) = 0$$

$$n * 2 \sum (-xy + mx^2 - c \cdot x) = 0$$

$$-\sum xy + \sum mx^2 - \sum c \cdot x = 0$$

$$\boxed{\sum mx^2 = \sum c \cdot x + \sum xy} \quad -(1)$$

$$\frac{\partial E}{\partial c} = \sum \frac{\partial E}{\partial c} \left(y - (mx + c)\right)^2 = 0$$

$$n * 2 \sum (y - mx - c) = 0$$

$$\sum y - \sum mx - \sum c = 0$$

↳ suming a constant over a limit is $n * c$

$$\sum y - \sum mx = n * c$$

$$c = \frac{\sum y}{n} - \frac{m \sum x}{n}$$

$\frac{\sum y}{n}$ → average of all y

$\frac{\sum x}{n}$ → average of all x

$$\boxed{c = \bar{y} - m\bar{x}} \quad -(2)$$

$\sum x^2$
$\sum xy$
$\sum x$
$\bar{x}$
$\bar{y}$

} can be computed using numpy and pandas.